AFFINE MORPHISMS ARE LIMITS OF AFFINE FP MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2924-2936, following the statement at 2916-2922.

Editorial note. The final paragraph is a construction-reading step ('By construction ...') anchored to relative Spec and the algebra engine; flagged in the notes.

Allowed standard prerequisites (editorial).

- Algebras as directed colimits of fp algebras. lemma-algebra-directed-colimit-finite-presentation (same chapter, cited at ~2926-2928). A = f_*O_X = colim A_i with A_i finitely presented O_Y-algebras.
- Relative Spec. Morphisms of Spaces, Definition spaces-morphisms-definition-relative-spec (cross-chapter, cited at ~2929-2931). X_i = relative Spec_Y(A_i).

Source reference. Original lemma 082A: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2580–2580. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 081W: focused-seeds/geometry/S9-spaces-limits-affine-morphism-is-limit/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be an affine morphism of algebraic spaces over S. If Y quasi-compact and quasi-separated, then X is a directed limit $X = \lim X_i$ with each X_i affine and of finite presentation over Y.*

Proof. Consider the quasi-coherent $\mathcal{O}_Y$-module $\mathcal{A} = f_*\mathcal{O}_X$. By Lemma 082A we can write $\mathcal{A} = \operatorname{colim} \mathcal{A}_i$ as a directed colimit of finitely presented $\mathcal{O}_Y$-algebras $\mathcal{A}_i$. Set $X_i = \underline{\operatorname{Spec}}_Y(\mathcal{A}_i)$, see Morphisms of Spaces, Definition 081W. By construction $X_i \to Y$ is affine and of finite presentation and $X = \lim X_i$. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.